Name: ________________________

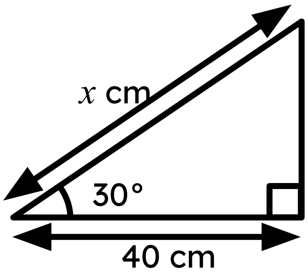
Worksheet

Choosing an appropriate method for findings lengths of a triangle

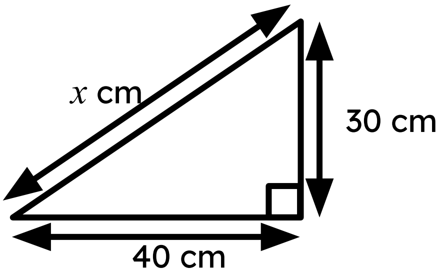
Task A: Trigonometry or not

1) Calculate x in each triangle.

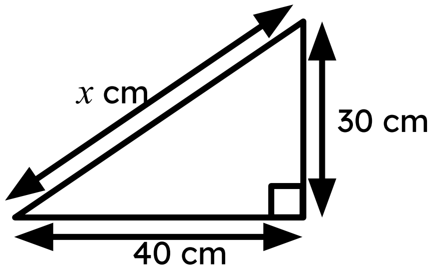
a)



b)

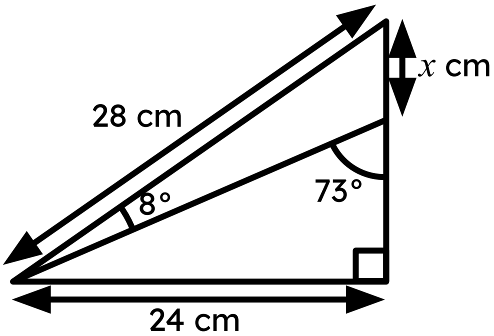


c) The perimeter is 1.2 m

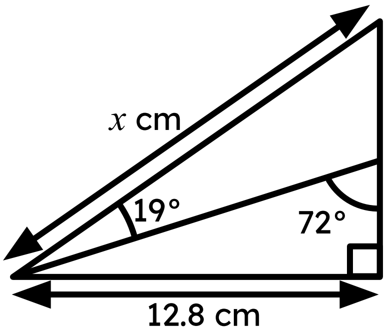


2) Calculate x in each triangle, giving your answers to 1 decimal place.

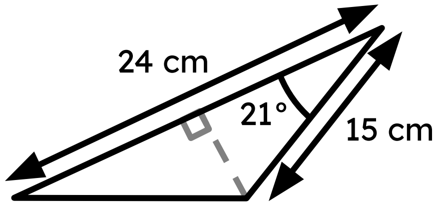
a)



b)



3) Work out the perimeter of this triangle, giving your answer to 1 decimal place.



Task B: Practical uses of trigonometry

1) A ramp is being built outside a house.
The bottom of the door is 42 cm off the floor.
The angle between the ramp and the ground should not exceed 4.8°
What is the minimum slope length of the ramp?

2) A man standing on a river-bank notes a particular tree on the opposite bank.
He then walks 60 metres along the river-bank in a straight line.
On looking back he can see that the angle between the line of his path and a line to the tree is 32°.
How wide is the river?

3) A 7 metre pole is secured by support wires. The wires are pegged in a straight line. The outer wires make an angle of 31° with the floor and the inner wires make an angle twice the size. How far apart are the wires?